KAIZEN MATHS

Lesson starter

Logarithms and Exponential Equations

Fundamentals of Logarithms

Name _____

S1

Write in index form: $\log_4(16) = 2$

S2

Write in log form: $5^2 = 25$

S3

Evaluate: $\log_3(9)$

S4

Write in index form: $\log_5(25) = 2$

Show your working

KAIZEN MATHS

Lesson starter

Logarithms and Exponential Equations

Fundamentals of Logarithms

Name _____

S1

Write in index form: $\log_4(16) = 2$

S2

Write in log form: $5^2 = 25$

S3

Evaluate: $\log_3(9)$

S4

Write in index form: $\log_5(25) = 2$

ALG-9558-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Logarithms and Exponential Equations

Fundamentals of Logarithms

Name _____

E1

Write in log form: $5^4 = 625$

E2

Evaluate: $\log_5(125)$

E3

Write in index form: $\log_5(625) = 4$

KAIZEN MATHS

Exit ticket

Logarithms and Exponential Equations

Fundamentals of Logarithms

Name _____

E1

Write in log form: $5^4 = 625$

E2

Evaluate: $\log_5(125)$

E3

Write in index form: $\log_5(625) = 4$

ALG-9558-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Logarithms and Exponential Equations

Fundamentals of Logarithms

Name _____

E1

Write in log form: $5^4 = 625$

E2

Evaluate: $\log_5(125)$

E3

Write in index form: $\log_5(625) = 4$

ALG-9558-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Logarithms and Exponential Equations

Fundamentals of Logarithms

Name _____

E1

Write in log form: $5^4 = 625$

E2

Evaluate: $\log_5(125)$

E3

Write in index form: $\log_5(625) = 4$

ALG-9558-L1 · E

Show your working

Teacher answers & working

Logarithms and Exponential Equations · Fundamentals of Logarithms

ALG-9558-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Write in index form: $\log_4(16) = 2$

Answer:

$4^2 = 16$

Working:

Method focus: A logarithm is another way of writing an index equation.

Identify the base, argument, and value

Base = 4, argument = 16, value = 2

Use the definition

$\log_a(x) = n$ means $a^n = x$

Apply it

$\log_4(16) = 2$ means $4^2 = 16$

$4^2 = 16$

Final answer: $4^2 = 16$

S2

Write in log form: $5^2 = 25$

Answer:

$\log_5(25) = 2$

Working:

Method focus: A logarithm is another way of writing an index equation.

Identify the base, power, and result

Base = 5, power = 2, result = 25

Use the definition

$a^n = x$ means $\log_a(x) = n$

Apply it

$5^2 = 25$ means $\log_5(25) = 2$

$\log_5(25) = 2$

Final answer: $\log_5(25) = 2$

Teacher answers & working

Logarithms and Exponential Equations · Fundamentals of Logarithms

ALG-9558-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Evaluate: $\log_3(9)$

Answer:

2

Working:

Method focus: Ask which power of the base gives the argument.

Ask what the logarithm means

$\log_3(9)$ asks: 3 to what power gives 9?

Write the matching index equation

$3^2 = 9$

Read off the power

$\log_3(9) = 2$

Final answer: 2

S4

Write in index form: $\log_5(25) = 2$

Answer:

$5^2 = 25$

Working:

Method focus: A logarithm is another way of writing an index equation.

Identify the base, argument, and value

Base = 5, argument = 25, value = 2

Use the definition

$\log_a(x) = n$ means $a^n = x$

Apply it

$\log_5(25) = 2$ means $5^2 = 25$

$5^2 = 25$

Final answer: $5^2 = 25$

Teacher answers & working

Logarithms and Exponential Equations · Fundamentals of Logarithms

ALG-9558-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Write in log form: $5^4 = 625$

Answer:

$\log_5(625) = 4$

Working:

Method focus: A logarithm is another way of writing an index equation.

Identify the base, power, and result

Base = 5, power = 4, result = 625

Use the definition

$a^n = x$ means $\log_a(x) = n$

Apply it

$5^4 = 625$ means $\log_5(625) = 4$

$\log_5(625) = 4$

Final answer: $\log_5(625) = 4$

E2

Evaluate: $\log_5(125)$

Answer:

3

Working:

Method focus: Ask which power of the base gives the argument.

Ask what the logarithm means

$\log_5(125)$ asks: 5 to what power gives 125?

Write the matching index equation

$5^3 = 125$

Read off the power

$\log_5(125) = 3$

Final answer: 3

Teacher answers & working

Logarithms and Exponential Equations · Fundamentals of Logarithms

ALG-9558-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Write in index form: $\log_5(625) = 4$

Answer:

$5^4 = 625$

Working:

Method focus: A logarithm is another way of writing an index equation.

Identify the base, argument, and value

Base = 5, argument = 625, value = 4

Use the definition

$\log_a(x) = n$ means $a^n = x$

Apply it

$\log_5(625) = 4$ means $5^4 = 625$

$5^4 = 625$

Final answer: $5^4 = 625$